

Dual ordinal spectral ranking of journals and institutions

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Abstract

We develop a dual spectral ranking of journals and institutions that jointly determines prestige scores via a Katz resolvent applied to a block attention matrix constructed from intra-field citation flows. A continuous parameter χ apportions each reference between source and institution channels; a binary block-mask selects among four bibliometrically distinct flow structures, nesting conventional single-mode rankings as special cases. Applied to the economics and commerce literature using OpenAlex data, the method produces stable, interpretable joint rankings whose sensitivity to model parameters is examined systematically.

Keywords: Scientometrics, spectral ranking, dual journal–institution prestige, Katz resolvent, Pinski–Narin

1 Introduction

Journal and institution ranking. The research system relies on journals and institutions. Journals filter works through editorial and peer review, archive the literature, and create metadata for retrieval. Institutions fund individual researchers, implement training, oversee compliance, and maintain physical and information infrastructure. [Placeholder: overview of the joint ranking problem.]

Previous work on journal ranking. ? introduced influence-per-reference weights for journal ranking; Vigna (2016) surveys the broader lineage; ? traces the Katz–Hubbell–Pinski–Narin–PageRank chain. [Placeholder: expand journal ranking literature.]

How institutions are ranked. Institutional rankings, dominated by QS, Times Higher Education (THE), and ShanghaiRanking (ARWU), blend bibliometrics with reputational surveys, teaching metrics, and internationalisation. The Leiden Ranking counters with field-normalised bibliometrics, revealing niche leaders but lacking the reputational component of QS and THE. Spectral methods have not previously been applied jointly to sources and institutions. [Placeholder: expand institutional ranking literature.]

Theoretical framing. Following Wouters (1999), a *reference* written by a scientist becomes a *citation*—a directed edge (w_i, w_j) —only after indexer resolution to canonical work records.

We treat citations as structural relations within a functionally differentiated scientific system (?), setting aside authorial intent. Dual spectral rankings of journals and institutions treat citations as autopoietic communications within science’s communicative system (Social System of Science Citation Theory, SSCT), where individual actors reside in the environment rather than constituting system elements (Tahamtan and Bornmann, 2022). This contrasts with actor-network theory (ANT), which foregrounds heterogeneous agents in assembling citation networks; ANT enters only in the Discussion. The resulting dual rankings are self-referential hierarchies that condition future citations. Our construction extends Breiger (1974): all four blocks of the joint attention matrix are primary flows, not derived projections.

Paper structure. Section 2 develops the mathematical framework. Section ?? describes the corpus construction and source selection. Section 4 presents results. Section 5 discusses validity and prospects.

2 Journal–institution citation networks

We begin corpus construction by selecting N_s sources $\mathfrak{S}$ in a field, discipline or category. Then we select N_w works $\mathfrak{W}$ published in $\mathfrak{S}$ within the census window years $[t_0^c, t_1^c]$ or the earlier contiguous target year t^t where $t^t = t_0^c - 1$ (we explain this choice in the discussion). From the reference lists of the set of census works $\mathfrak{W}^c$ we identify the set of target works $\mathfrak{W}^t$ and form the filtered reference set $\mathfrak{R} = \{(i, j) : w_i \in \mathfrak{W}^c, w_j \in \mathfrak{W}^t\}$. Author affiliations in $\mathfrak{W}$ establish a set of institutions $\mathfrak{U}'$ and a set of authors $\mathfrak{A}'$. Examining the frequency distributions of institutions and authors, we retain institutions with work counts $\geq \tau^U$ and authors with work counts $\geq \tau^A$, giving $\mathfrak{U}$ and $\mathfrak{A}$ respectively. Each work w_i has an *institution weight* under author-fractional counting (Waltman and van Eck, 2015),

$$\omega_{iu} = \sum_{\ell: u \in \Pi_{i\ell}} \frac{1}{a_i} \frac{1}{u_{i\ell}}, \quad (1)$$

where ℓ indexes authors of work w_i , a_i is the number of authors, and $u_{i\ell}$ the number of institutions of author ℓ . Since ω_{iu} requires both an author and an institution, we drop any authorship where either the author $\notin \mathfrak{A}$ or the institution $\notin \mathfrak{U}$, and subsequently drop works that lose all their authorships. The institution weights are not renormalised to account for the omission of an authorship, so that $\sum_u \omega_{iu} \leq 1$.

When a work w_i references another work w_j we say that w_i gives attention to w_j . The attention given by a work can be either (a) the number of references, or (b) a fixed amount (Zitt and Small, 2008). Let $R_i = |\{j : (i, j) \in \mathfrak{R}\}|$ be the reference count of work w_i and $\bar{R} = |\mathfrak{R}|/N_w$ the mean reference count over $\mathfrak{W}$. We use the *reference normalisation weight* δ_i to select case (a) or case (b) respectively:

$$\delta_i = \begin{cases} 1 & \text{(full count)} \\ \bar{R}/R_i & \text{(fixed count).} \end{cases} \quad (2)$$

We construct the set of *units* [sources or institutions; Pinski and Narin (1976)] $\mathfrak{P} = \mathfrak{S} \cup \mathfrak{U}$ with $N = N_s + N_u$ elements, ordered so that sources occupy indices $1, \dots, N_s$ and institutions indices $N_s + 1, \dots, N$. A reference $(i, j) \in \mathfrak{R}$ gives attention across unit-type pairs: source–source (*SS*), source–institution (*SI*), institution–source (*IS*), and institution–institution (*II*).

For each work w_i we define the source membership one-hot vector $\mathbf{b}_i^S \in \mathbb{R}^{N_s}$ with $[\mathbf{b}_i^S]_s = \mathbf{1}[w_i \in s]$, and the institution weight distribution vector $\mathbf{b}_i^U \in \mathbb{R}^{N_u}$ with $[\mathbf{b}_i^U]_u = \omega_{iu}$ from (1). Attention flow between pairs of unit types is accumulated as four blocks over $\mathfrak{R}$ as rank-1 outer products,

$$\mathbf{C}_{XY} = \sum_{(i,j) \in \mathfrak{R}} \delta_i \mathbf{b}_i^X (\mathbf{b}_j^Y)^\top, \quad XY \in \{SS, SI, IS, II\}, \quad (3)$$

with $\mathbf{b}_i^X$ weighting the referencing (citing) side and $\mathbf{b}_j^Y$ the referenced (cited) side.

The citation matrix assembles the four blocks:

$$\mathbf{C}(\chi, \mathbf{m}, \delta) = \begin{pmatrix} m_{SS}(1-\chi)^2 \mathbf{C}_{SS} & m_{SI}\chi(1-\chi) \mathbf{C}_{SI} \\ m_{IS}\chi(1-\chi) \mathbf{C}_{IS} & m_{II}\chi^2 \mathbf{C}_{II} \end{pmatrix}, \quad (4)$$

where the binary *block mask* $\mathbf{m} = (m_{SS}, m_{SI}, m_{IS}, m_{II}) \in \{0, 1\}^4$ is bibliometrically meaningful for source-only $(1, 0, 0, 0)$, institution-only $(0, 0, 0, 1)$, bipartite $(0, 1, 1, 0)$, and full-joint $(1, 1, 1, 1)$ combined blocks. The mixing weight χ controls the source–institution balance in the bipartite and full-joint cases and is inert in the single-mode cases. The χ scaling preserves total citation flow across all block combinations, since $(1-\chi)^2 + 2\chi(1-\chi) + \chi^2 = 1$. We could name $\mathbf{C}$ the reference, attention or citation matrix and prefer the latter.

The model parameters are summarised in Table 1. Note that the corpus excludes (a) census work references not among target works, (b) non-census works that reference target works, and (c) authorships that include an omitted author or institution.

Table 1: Model parameters θ .

Symbol	Name	Type	Role
<i>Corpus construction</i>			
$\mathfrak{S}$	source set	set	define the field, subject, category
$[t_0^c, t_1^c]$	census window	interval	works whose references are traced
$t^t = t_0^c - 1$	target year	integer	the year immediately preceding the census window; works eligible to receive references
τ^U	institution threshold	integer	minimum work count to retain an institution
τ^A	author threshold	integer	minimum work count to retain an author
<i>Matrix construction</i>			
$\delta \in \{0, 1\}$	reference normalisation	binary	0: a work contributes $\bar{R}$ units; 1: a reference contributes 1 unit
$\mathbf{m} \in \{0, 1\}^4$	block mask	binary 4-vector	select active unit-type pairs (SS , SI , IS , II)
$\chi \in [0, 1]$	mixing weight	continuous	source-institution balance within each work
<i>Ranking</i>			
$\alpha \in [0, 1)$	damping factor	continuous	controls propagation through citation network; ensures convergence

Spectral ranking. Let $r_p = \sum_q C_{pq}$ be the total outgoing citation flow from unit p , $\mathbf{D}_r = \text{diag}(r_p)$ the corresponding diagonal matrix. If $r_p = 0$ (a dangling unit with no outgoing citations) the corresponding row of $\mathbf{C}$ is zero and $\mathbf{H}$ is defined to have a zero row there; the boundary condition below then supplies the necessary probability mass. Let $\boldsymbol{\mu}$ be an initial preference, or boundary condition (Hubbell, 1965; Vigna, 2016), with entries $\mu_p = a_p / \sum_q a_q$, where the unit work count a_p is $\Sigma_s = |\{i : w_i \in s\}|$ for source $p = s$ and $\Sigma_u = \sum_i \omega_{iu}$ for institution $p = u$, the latter being the fractional work count summed via (1). The row-stochastic citation matrix is $\mathbf{H} = \mathbf{D}_r^{-1} \mathbf{C}$.

The spectral ranking $\boldsymbol{\pi}$ (see Vigna, 2016, for an overview) is defined as the solution to the damped eigenvector problem

$$\boldsymbol{\pi} = \alpha \mathbf{H}^\top \boldsymbol{\pi} + (1 - \alpha) \boldsymbol{\mu}, \quad \mathbf{1}^\top \boldsymbol{\pi} = 1, \quad \boldsymbol{\pi} \geq 0, \quad (5)$$

with damping factor $\alpha \in [0, 1)$. The prior $\boldsymbol{\mu}$ encodes the distribution over units weighted by scholarly output, and α controls the degree to which influence propagates through the citation network rather than being absorbed by that prior. In the limit $\alpha \rightarrow 1$, equation (5) reduces to the principal eigenvector problem

$$\boldsymbol{\pi}^\infty = \mathbf{H}^\top \boldsymbol{\pi}^\infty, \quad \mathbf{1}^\top \boldsymbol{\pi}^\infty = 1, \quad \boldsymbol{\pi}^\infty \geq 0, \quad (6)$$

whose solution is unique when $\mathbf{H}$ is primitive (irreducible and aperiodic). Geller (1978) shows that $\boldsymbol{\pi}^\infty$ is then the stationary distribution of the Markov chain with transition matrix $\mathbf{H}$, and that it coincides with the *influence weight* of Pinski and Narin (1976) scaled by the reference count proportion $r_p / \sum_q r_q$.

The spectral ranking $\boldsymbol{\pi}$ reflects the total citation flow received by each unit and therefore depends on unit size. *Prestige per work* (Palacios-Huerta and Volij, 2004),

$$\mathbf{v}(\alpha) = \mathbf{A}^{-1} \boldsymbol{\pi}(\alpha), \quad \mathbf{A} = \text{diag}(a_p), \quad (7)$$

removes this dependence, where $\boldsymbol{\pi}(\alpha)$ denotes the solution to (5) and $\boldsymbol{\pi}^\infty$ the limiting case (6). Palacios-Huerta and Volij (2004) show that $\mathbf{v}$ is uniquely characterised by several reasonable axiomatic constraints (but see Serrano, 2004). Note that $\mathbf{A}^{-1}$ must be applied after convergence; applying it inside the iteration loop would iterate on $\mathbf{H}^\top \mathbf{A}$, which is not norm-preserving and yields a fixed point with no clean interpretation.

Computation via power iteration. To compute $\boldsymbol{\pi}$ we define the modified matrix

$$\tilde{\mathbf{H}} = \alpha \mathbf{H} + (1 - \alpha) \boldsymbol{\mu} \mathbf{1}^\top, \quad (8)$$

so that the fixed-point condition $\boldsymbol{\pi} = \tilde{\mathbf{H}}^\top \boldsymbol{\pi}$ is equivalent to equation (5). Because $\boldsymbol{\mu} > 0$ and $0 < \alpha < 1$, every entry of $\tilde{\mathbf{H}}$ is strictly positive, ensuring it is a primitive row-stochastic matrix. By the Perron–Frobenius theorem, $\tilde{\mathbf{H}}^\top$ therefore has a unique stationary distribution $\boldsymbol{\pi}$, recovered by power iteration

$$\boldsymbol{\pi}^{(k+1)} = \tilde{\mathbf{H}}^\top \boldsymbol{\pi}^{(k)}, \quad (9)$$

converging to $\boldsymbol{\pi}$ from any strictly positive starting vector $\boldsymbol{\pi}^{(0)}$.

Computation. $\mathbf{C}(\chi, \mathbf{m}, \delta)$ is a sparse matrix built from a weighted edge list $\mathcal{E}$ accumulated over $\mathfrak{R}$ with per-reference weight δ_i , stored in CSR format, and row-normalised to $\mathbf{H}$. The pipeline $\mathfrak{R} \rightarrow \mathcal{E} \rightarrow \mathbf{C} \rightarrow \mathbf{H} \rightarrow \boldsymbol{\pi} \rightarrow \mathbf{v}$ is implemented using `scipy.sparse.csr_matrix` with parameter entry points detailed in the Supplement.

3 Corpus selection

Corpus extraction starts with a list of eligible sources. We require each source to have three features: an OpenAlex source identifier for corpus extraction, evidence from one of our journal registries that the source publishes economics or business research, and high OpenAlex source topic density to exclude cross-disciplinary journals and other edge cases.

The journal registries are as follows. Harzing’s *71st Edition of the Journal Quality List* (Harzing, 2024, JQL) assembles peer assessments of journal quality across business and economics disciplines. Its scope is already confined to the target fields and no filtering is applied. The *ERA 2023 Submission Journal List* (Australian Research Council, 2022, SJL) assigns up to three Field of Research (FoR) codes to each entry. From the full list of 26,241 journals we retain those having a FoR code falling within Division 35 (Commerce, Management, Tourism and Services) or Division 38 (Economics). This applies to two-digit ‘division’ codes and all four-digit ‘group’ codes (3501–3509, 3801–3803, 3899). After filtering, 2,469 journals remain. *Clarivate Web of Science Journal Citation Reports* (Clarivate Analytics, 2026, JCR) assign each journal a Web of Science category. We retain those in Economics, Management, Business, Transportation or Business, Finance. After filtering, 1,413 journals remain.

Each journal in each of JQL, SJL, and MJL have at least one ISSN, and may have several. These registry ISSNs are joined independently to the OpenAlex sources snapshot using its list of ISSNs as the matching key. Journal titles are examined to identify around 40 additional matches where the OpenAlex source does not have an ISSN. The list of OpenAlex sources that are matched is designated OAS*. matching outcomes are summarised in Table 2.

The long list OAS* includes cross-disciplinary sources publishing works in disciplines beyond economics and business, or otherwise not scholarly works (e.g. newsletters labeled as journals). We use OpenAlex source-topic assignments to exclude sources having a minority of works in economics/business. A source has work-topic counts for up to five topics, each within a single sub-field, field, and domain. We accumulate the work-topic counts for Field 14 (Economics, Econometrics and Finance) and Field 20 (Business and Management) and define the work-topic density is the ratio of this count to the total works in the source. Sources are dropped if this ratio is below 50%. The 2,289 surviving sources, labeled OAS, are used to extract the corpus. Matching statistics for the three registries and a summary of reasons for excluding sources are shown in Table 2.

4 Results

Baseline model. [Placeholder.]

Parameter sensitivity. [Placeholder: $\alpha \in \{0.50, 0.75, 0.85, 0.95\}$; $\chi \in \{0.00, 0.25, 0.50, 0.75, 1.00\}$; $\mathbf{m} \in \{(1, 0, 0, 0), (0, 1, 1, 0), (1, 1, 1, 1)\}$; $\rho \in \{0, 1\}$; τ sweep.]

Bootstraps. [Placeholder.]

Table 2: Matching of the three register-based sources to OpenAlex source identifiers. Pairwise overlaps are based on shared among matched entries.

Register	Journals	Matched	Match rate	Unmatched
JQL	827	826	99.9%	1
MJL	1,413	1,393	98.6%	20
SJL	2,469	2,193	88.8%	276
<i>Pairwise overlap of matched OAS</i>				
MJL $\cap$ SJL	1,008 sources			
MJL $\cap$ JQL	540 sources			
SJL $\cap$ JQL	618 sources			
MJL $\cap$ SJL $\cap$ JQL	505 sources			
Union (long-list)	2,751 unique OAS			

5 Discussion

Face validity. [Placeholder.]

Content validity. [Placeholder.]

Criterion validity — concurrent and predictive. [Placeholder.]

Construct validity. [Placeholder.]

6 Prospects

[Placeholder.]

7 Prospects

[Placeholder.]

8 Addenda

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